

Network Diagram

Physical Network and Cybersecurity Topology

Network Diagram Assignment Task Submission

Prepared By:

Noe Valencia

Dallas College | Cyber Security Management

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Professional Experience and Summary

I have been working on network and cybersecurity projects throughout this program and have built a solid foundation in designing and evaluating secure network environments. My focus has been on practical application, which is why this project was built around a real network topology rather than a hypothetical one. I used the actual Dallas Office network as the starting point and worked through the full assessment and redesign process.

Network Systems Architecture

I have hands-on experience designing and analyzing unified network infrastructures. This includes working with firewall configurations, VLAN segmentation, structured cabling, and server environments. For this project specifically, I analyzed the existing topology, ran a full CSET assessment against NIST CSF 2.0, identified the gaps, and then built out the upgraded design to address each finding directly. The goal was not to just recommend changes but to show exactly what those changes look like in practice.

Cybersecurity

My cybersecurity work in this program has covered assessment tools, policy development, incident response planning, risk management, and physical security design. The CSET assessment I ran on the Dallas Office network came back with a Security Assurance Level of Low. The top five areas of concern were network monitoring, incident response, personnel training, technology usage monitoring, and threat intelligence. Every upgrade in this design ties back to one of those findings.

Servers	Dell R640, MD1400, BE6K	4x Dell PowerEdge R750 with Veeam Backup
Cabling	CM-rated Cat5	Cat6a plenum rated
VoIP	Cisco BE6K, no secure gateway	VOIP GW01 on Virtual Wire through PA-5220
IDS/IPS and monitoring	None	Palo Alto IDS/IPS and Splunk SIEM
Identity and MFA	None	Okta SSO/MFA, CyberArk PAM, YubiKey
Endpoint protection	None	Huntress Managed EDR and MS Defender E5
Cameras	11x analog with local DVR	24x Verkada CD62 4K IP cameras, cloud managed
Door access	8x numeric keypad only	16x Verkada card readers with HID proximity cards
Power backup	None	2x Eaton 9PX 10kVA UPS and emergency generator

Physical and Cybersecurity Design

The security plan for this network covers two things. The first is physical security, which means controlling who can get into the building and into the rooms where the network equipment lives. The second is cybersecurity, which covers how the network itself is protected. Both were evaluated using the CSET tool against NIST CSF 2.0 and both had significant gaps that needed to be addressed.

The LMEC example proposal that was used as a reference for this course describes physical and cybersecurity as interdependent. A failure in one can lead to a failure in the other. That is the approach I took here as well. You cannot just add a better firewall and call it done if someone can still walk into the server room without any verification.

Physical Security

The current setup has analog cameras recording to a local DVR and numeric keypads on the doors. There is no way to verify who entered a room or review footage remotely if the DVR is damaged or stolen. Here is what the upgraded design adds:

- The analog cameras get replaced with 24 Verkada CD62 4K IP cameras. These are cloud managed, which means footage is backed up automatically and can be reviewed from anywhere. They have motion detection and night vision built in. Cameras are placed at all entry points, in both hallways, inside the MDF closet, and inside the server room.
- The numeric keypads get replaced with 16 Verkada card readers using HID Global proximity cards. A card reader goes on every door that leads to a restricted area. That includes the MDF room, the server room, the IT workspace, the staff corridor entrances, and the main building entry. The system logs every access event so there is always a record of who entered and when.
- Biometric fingerprint safes are added inside the server rooms for securing physical media and backup drives. Four safes are included in the budget.
- Two Eaton 9PX 10kVA rack-mounted UPS units keep the network running during a power outage. An emergency generator provides longer-term backup. This means the cameras, card readers, switches, and servers all stay online even if the building loses power.

Cybersecurity

The CSET assessment ranked the top concerns in order. Network monitoring was number one. Incident response was number two. Training was three and four. Technology usage monitoring was five. The upgrades below are listed in that order because those are the priorities the assessment identified.

- The Palo Alto PA-5220 NGFW running as an HA pair replaces the Adtran 908. This addresses the number one finding. The PA-5220 does deep packet inspection, runs IDS/IPS, handles NAT, and routes VOIP traffic through a Virtual Wire connection so phone calls go through the firewall perimeter. Running two units in a high-availability pair means if one fails the other takes over automatically.
- Splunk Enterprise SIEM with 250GB per day ingest gives the team a central place to see what is happening across the entire network. Logs come in from the firewall, switches, servers, endpoints, and authentication systems. This is the direct fix for DE.CM-01, which was the highest ranked gap.
- Zscaler Zero Trust Exchange handles secure remote access for all 200 users. Instead of a traditional VPN, every connection gets evaluated before access is granted. This closes the remote access gap that the original design had no answer for.
- The Ubiquiti UniFi APs get replaced with 12 Cisco Meraki MR57 Wi-Fi 6E access points. The Staff SSID requires 802.1X domain authentication before any device connects to internal resources. The Public SSID is isolated in its own VLAN and only has internet access. A guest or attacker on the Public network cannot reach anything internal.

- Okta handles single sign-on and MFA for all 200 users. CyberArk manages privileged accounts so admin credentials are vaulted and not floating around. YubiKey 5 NFC hardware tokens are used for administrative logins so even if a password gets stolen there is no way to log in without the physical key.
- Huntress Managed EDR sits on top of Microsoft Defender for Endpoint E5 on all 200 endpoints. Huntress provides 24/7 SOC coverage which means someone is always watching the alerts even after hours. This addresses DE.CM-03 which flagged that personnel activity and technology usage were not being monitored.
- VLANs are used to separate traffic into four zones. VLAN 1 covers general endpoints at 192.168.0.0/16. VLAN 2 covers administration at 172.16.0.0/16. VLAN 15 covers VoIP phones at 10.10.6.0/24. VLAN 20 covers servers at 10.10.7.0/24. If something gets breached in one zone it cannot move freely into the others.
- Cat5 gets replaced with Cat6a plenum-rated cable throughout the building. Cat6a supports 10Gbps and meets fire code for air-handling spaces. The old Cat5 was neither.

NIST CSF 2.0 Control Mapping

The table below maps each of the top CSET findings to the control number and the specific upgrade that addresses it.

Control	CSET Finding	Upgrade Applied
DE.CM-01	No network monitoring for adverse events	Palo Alto PA-5220 IDS/IPS and Splunk SIEM
RC.RP-01	No incident response or recovery plan	Veeam Backup and documented IR procedures
PR.AT-02	No specialized role security training	KnowBe4 training and certification budget
PR.AT-01	No general personnel security training	KnowBe4 annual training for 200 users
DE.CM-03	No monitoring of personnel or technology usage	Splunk SIEM and Huntress Managed EDR
PR.AC-01	No 802.1X wireless authentication	Cisco Meraki MR57 802.1X and Okta MFA
PR.AC-06	No privileged access management	CyberArk PAM and YubiKey FIDO2
ID.AM-02	No software or systems inventory	Qualys VMDR automated asset discovery
ID.RA-01	No vulnerability identification or validation	Qualys VMDR and annual penetration testing